

Jenkins Task-2

Task Description:

Create a simple script file and push it to repo. Create a project in Jenkins connected to your GitHub repository. When a commit is made to your repo, automatically build must get triggered from Jenkins and the output must be shared to me via email.

Solution:

✓ 1. Launched EC2 and Installed Jenkins

```
ubuntu@ip-172-31-43-73:~$ sudo systemctl start jenkins
sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
ubuntu@ip-172-31-43-73:~$ sudo systemctl status jenkins
• jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-07-22 18:16:18 UTC; 16s ago
     Main PID: 5111 (java)
       Tasks: 51 (limit: 4670)
      Memory: 822.7M (peak: 841.9M)
         CPU: 16.112s
        CGroup: /system.slice/jenkins.service
                └─5111 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jul 22 18:16:13 ip-172-31-43-73 jenkins[5111]: aa235b7e5c1d46328572ff1860536f06
Jul 22 18:16:13 ip-172-31-43-73 jenkins[5111]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jul 22 18:16:13 ip-172-31-43-73 jenkins[5111]: *****
Jul 22 18:16:13 ip-172-31-43-73 jenkins[5111]: *****
Jul 22 18:16:13 ip-172-31-43-73 jenkins[5111]: *****
Jul 22 18:16:18 ip-172-31-43-73 jenkins[5111]: 2025-07-22 18:16:18.881+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
Jul 22 18:16:18 ip-172-31-43-73 jenkins[5111]: 2025-07-22 18:16:18.893+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Jul 22 18:16:18 ip-172-31-43-73 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jul 22 18:16:20 ip-172-31-43-73 jenkins[5111]: 2025-07-22 18:16:20.454+0000 [id=49] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data
Jul 22 18:16:20 ip-172-31-43-73 jenkins[5111]: 2025-07-22 18:16:20.454+0000 [id=49] INFO hudson.util.Retrier#start: Performed the action check updates s
lines 1-20/20 (END)
```

i-Oc10815ac99823d88 (Jenkins Task-2)
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- Need a remote Ubuntu server (EC2) to install Jenkins (CI/CD tool).

✓ 2. Create and Push Script to GitHub

```
ubuntu@ip-172-31-43-73:~$ nano hello.sh
ubuntu@ip-172-31-43-73:~$ cat hello.sh
#!/bin/bash
echo "Hello from Jenkins Auto-Build Job!"
ubuntu@ip-172-31-43-73:~$ chmod +x hello.sh
ubuntu@ip-172-31-43-73:~$
```

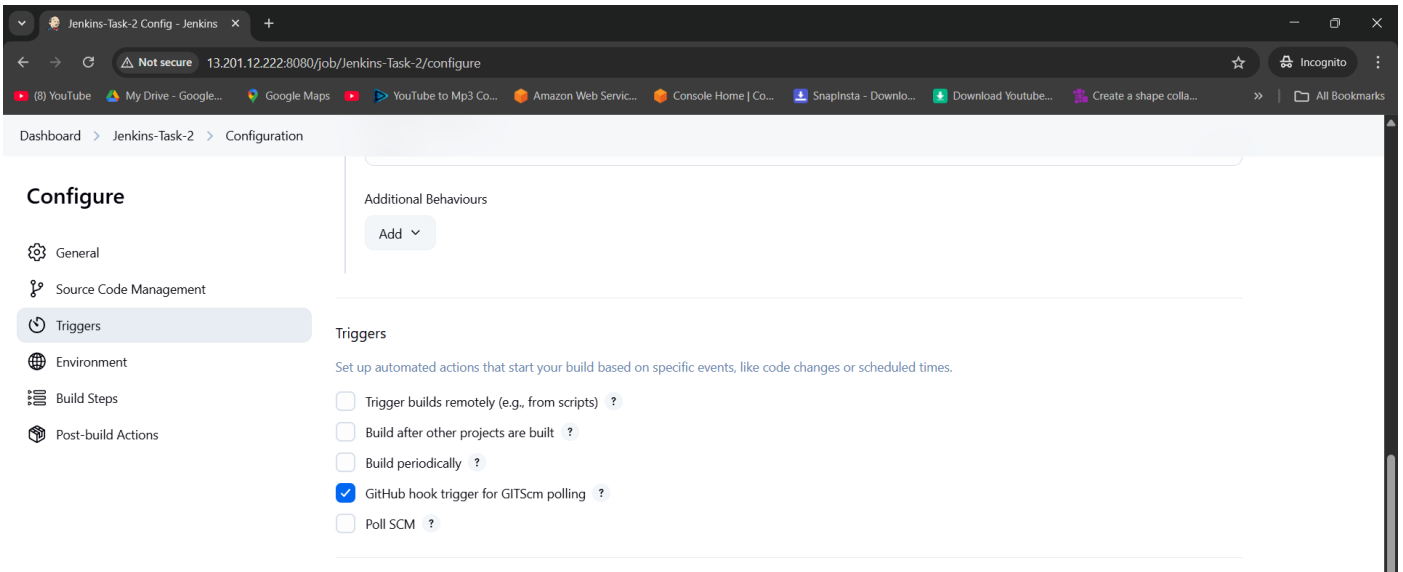
i-Oc10815ac99823d88 (Jenkins Task-2)
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```
ubuntu@ip-172-31-43-73:~$ git push -u origin master
Username for 'https://github.com': vigneshmarimuthu
Password for 'https://vigneshmarimuthu@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 297 bytes | 297.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/vigneshmarimuthu/jenkins-task-2.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
ubuntu@ip-172-31-43-73:~$
```

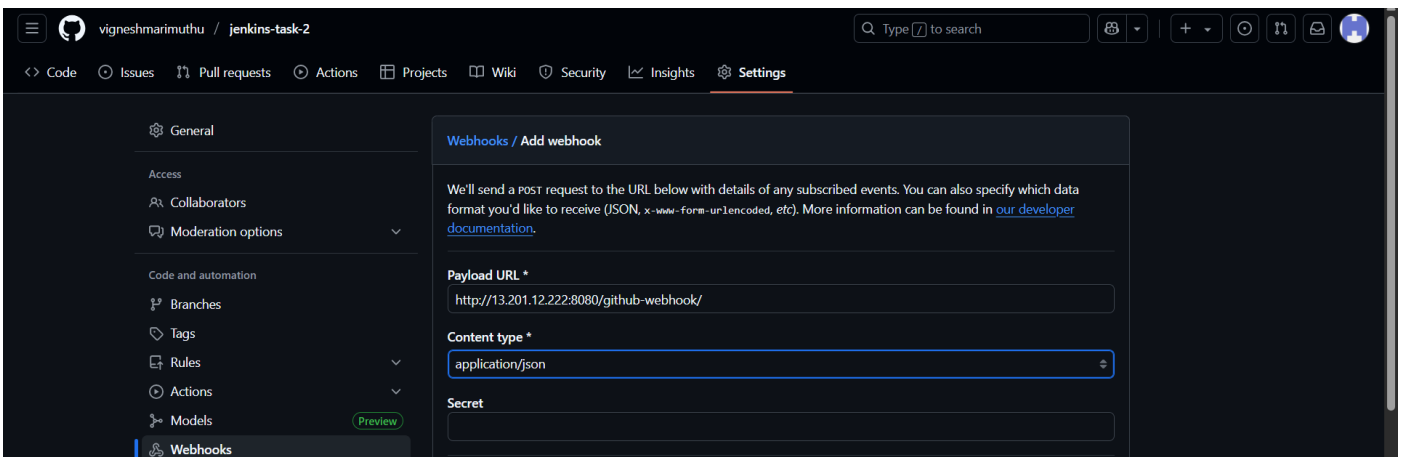
i-Oc10815ac99823d88 (Jenkins Task-2)
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- This script is what Jenkins will run when triggered.

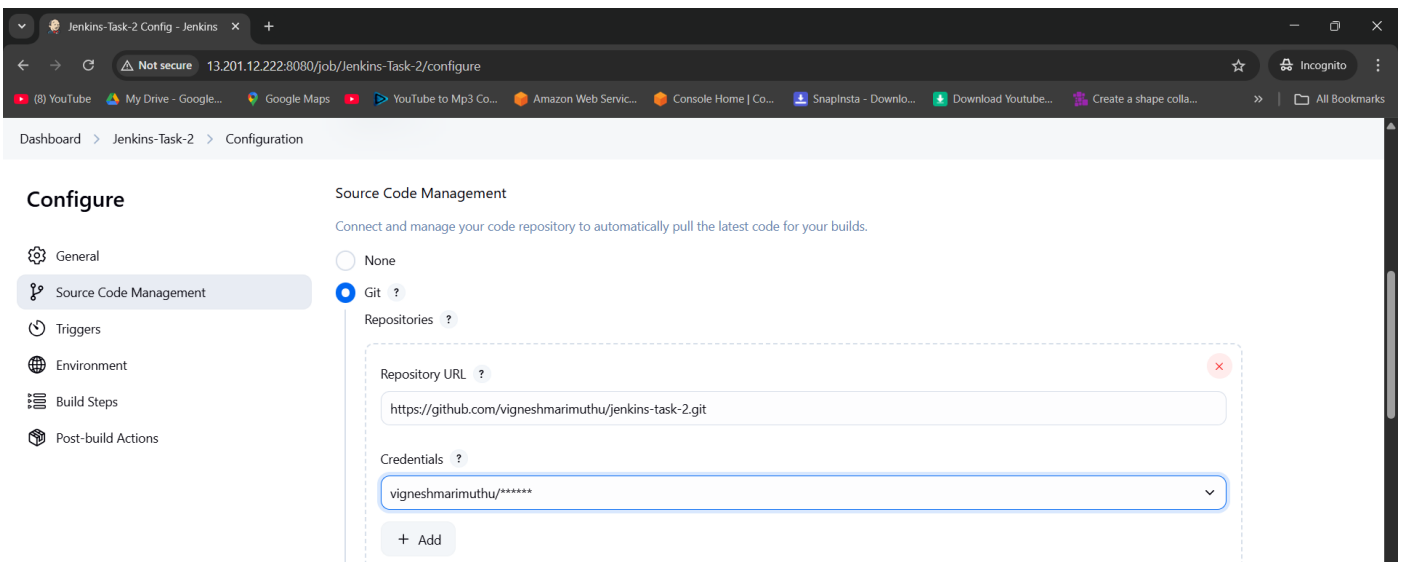
✓ 3. Configure GitHub Webhook and Jenkins Job



The screenshot shows the Jenkins Configuration page for 'Jenkins-Task-2'. The left sidebar lists configuration categories: General, Source Code Management, Triggers (selected), Environment, Build Steps, and Post-build Actions. The main content area is titled 'Configure' and shows the 'Triggers' section. Under 'Additional Behaviours', there is an 'Add' button. The 'Triggers' section includes a description: 'Set up automated actions that start your build based on specific events, like code changes or scheduled times.' Below this, there are several checkboxes: 'Trigger builds remotely (e.g., from scripts)', 'Build after other projects are built', 'Build periodically', 'GitHub hook trigger for GITScm polling' (which is checked), and 'Poll SCM'.



The screenshot shows the GitHub repository settings for 'jenkins-task-2'. The left sidebar lists various settings: General, Access, Collaborators, Moderation options, Code and automation, Branches, Tags, Rules, Actions, Models, and Webhooks (selected). The main content area is titled 'Webhooks / Add webhook'. It includes a description: 'We'll send a post request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, x-www-form-urlencoded, etc). More information can be found in our developer documentation.' Below this, there are fields for 'Payload URL' (http://13.201.12.222:8080/github-webhook/), 'Content type' (application/json), and 'Secret'.



The screenshot shows the Jenkins Configuration page for 'Jenkins-Task-2'. The left sidebar lists configuration categories: General, Source Code Management (selected), Triggers, Environment, Build Steps, and Post-build Actions. The main content area is titled 'Configure' and shows the 'Source Code Management' section. It includes a description: 'Connect and manage your code repository to automatically pull the latest code for your builds.' Below this, there are radio buttons for 'None' and 'Git' (which is selected). Under 'Repositories', there is a 'Repository URL' field (https://github.com/vigneshmarimuthu/jenkins-task-2.git) and a 'Credentials' dropdown menu (vigneshmarimuthu/*****). There is also an 'Add' button.

Jenkins

Dashboard > Jenkins-Task-2 > #1 > Console Output

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Next Build

Console Output

```
Started by user admin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Jenkins-Task-2
The recommended git tool is: NONE
using credential e41ee873-2561-40ec-a5d1-de9278dd3340
Cloning the remote Git repository
Cloning repository https://github.com/vigneshmarimuthu/jenkins-task-2.git
> git init /var/lib/jenkins/workspace/Jenkins-Task-2 # timeout=10
Fetching upstream changes from https://github.com/vigneshmarimuthu/jenkins-task-2.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/vigneshmarimuthu/jenkins-task-2.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
```

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- A webhook is a trigger. When code is pushed, GitHub tells Jenkins to build.

✓ 4. Configure Email Notifications

Jenkins-Task-2 Config - Jenkins

Dashboard > Jenkins-Task-2 > Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Editable Email Notification

Allows the user to disable the publisher, while maintaining the settings

☐ Disable Extended Email Publisher

Project From

Project Recipient List

Comma-separated list of email address that should receive notifications for this project.

vigneshmarimuthu1998@gmail.com

Project Reply-To List

Comma-separated list of email address that should be in the Reply-To header for this project.

\$DEFAULT_REPLYTO

Advanced

Edited

☒ Use SMTP Authentication

User Name

vigneshmarimuthu1998@gmail.com

Password

Concealed

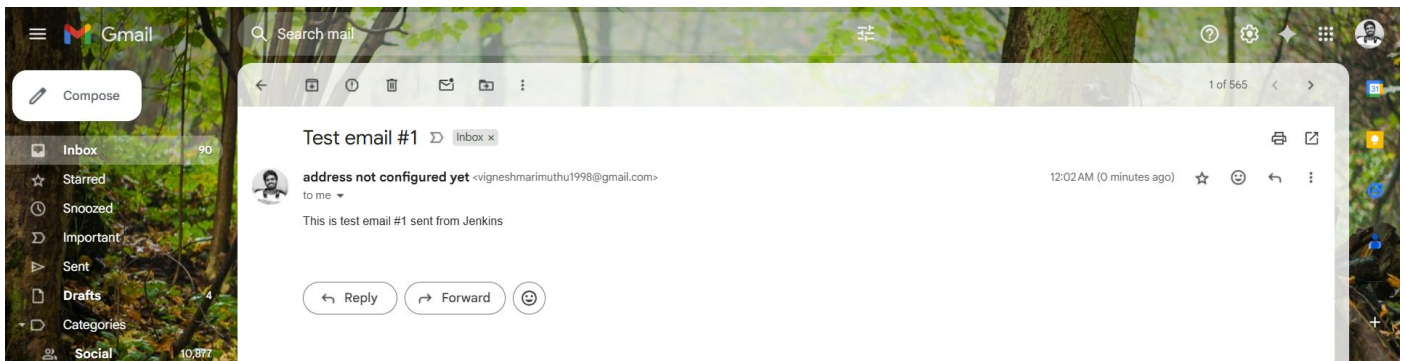
Change Password

☒ Use SSL

☐ Use TLS

SMTP Port

Save Apply



- Jenkins uses SMTP to send email alerts for build results.

✓ 5. Test Full Setup

```
ubuntu@ip-172-31-43-73:~$ git remote add origin https://github.com/vigneshmarimuthu/jenkins-task-2.git
ubuntu@ip-172-31-43-73:~$ git add hello.sh
ubuntu@ip-172-31-43-73:~$ git commit -m "Initial commit: added hello.sh"
[master (root-commit) 24ece61] Initial commit: added hello.sh
Committer: Ubuntu <ubuntu@ip-172-31-43-73.ap-south-1.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

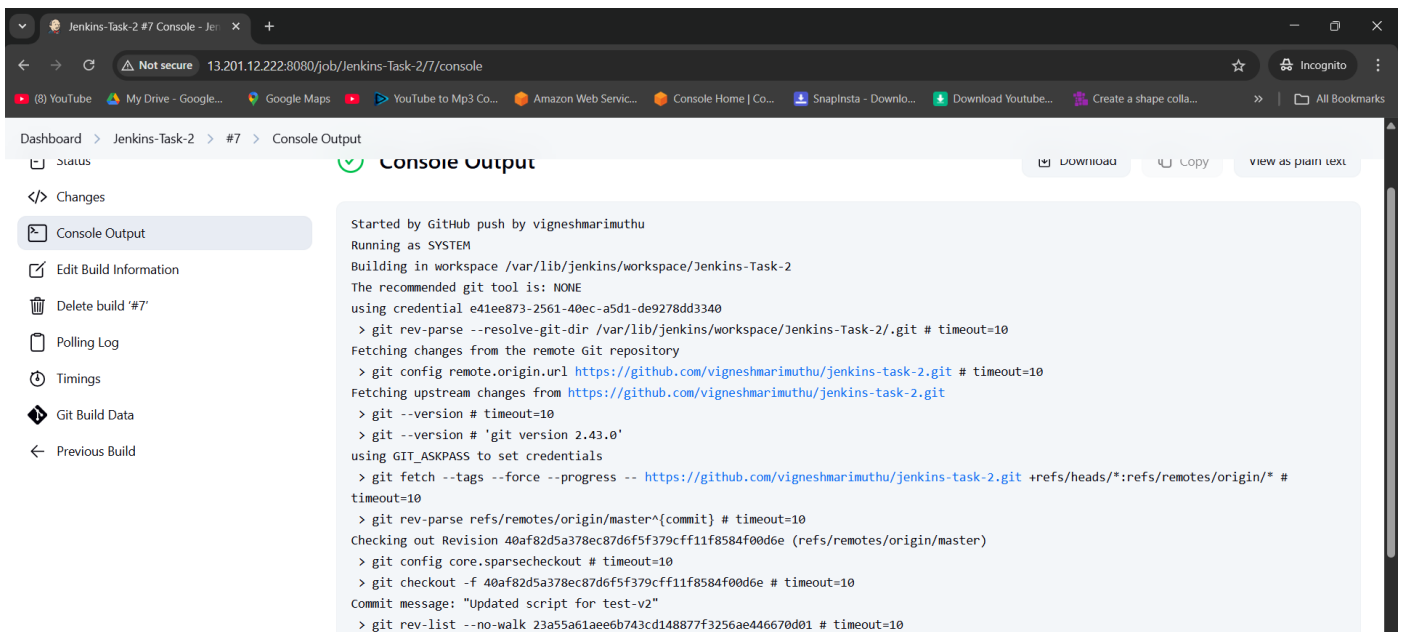
After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 2 insertions(+)
create mode 100755 hello.sh
ubuntu@ip-172-31-43-73:~$
```

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- Pushing a commit triggers Jenkins → script runs → email sent.